

JONATHAN LEVI, MD

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Executive Summary

Insight-driven **Data Scientist** with an M.D. and a strong foundation in mathematics, statistical modeling, and machine learning. Over two decades of experience in clinical practice and healthcare analytics, with a recent transition into advanced data science. Proven ability to derive actionable insights from complex datasets using Python, SQL, and data visualization tools. Passionate about leveraging predictive modeling and AI solutions to improve healthcare outcomes. Skilled in cross-functional collaboration, model optimization, and the integration of medical domain knowledge with data engineering to drive business value.

Skills

- Data Analysis
- Predictive Modeling
- Machine Learning
- Healthcare Analytics
- Statistical Computing
- Python Programming
- SQL Development
- Data Wrangling
- NLP Applications
- Risk Stratification
- Visualization Tools
- Time Series
- Clinical Informatics
- Feature Engineering
- Data Interpretation
- Performance Metrics
- Workflow Automation
- Research Methodology
- Deep Learning
- Big Data
- Model Evaluation

Professional Experience

Data Scientist | Software & Technical Consultant

05/2021 to Current

FREELANCE

- Engineered scalable machine learning models for predictive analysis of scientific and transportation data.
- Automated data pipelines for preprocessing and transformation of large, heterogeneous datasets.
- Modernized legacy systems by reducing redundancy, improving computational efficiency by 40%.
- Developed interactive dashboards using Python (Matplotlib, Plotly) to present insights to technical and non-technical stakeholders.
- Applied anomaly detection techniques to identify early indicators of disease from healthcare datasets.

Physician | Medical Data Analyst

06/1983 to 05/2021

Honolulu, HI; Kalamazoo, MI

- Conducted advanced statistical analyses of large patient cohorts to uncover predictive markers for cardiovascular disease.
- Leveraged machine learning to improve early diagnosis accuracy in clinical settings.
- Built custom Python tools for automating electronic health record (EHR) extraction and transformation.
- Championed AI adoption in diagnostics through integration of medical imaging and deep learning.
- Led cross-functional clinical research teams focused on applying data science to public health outcomes.

Educational Qualifications

Bachelor of Arts: Mathematics

Columbia University

MD

New York University School of Medicine

Projects Accomplished

- Glucose Management System, https://github.com/drjlevi6/Data-Scientist/tree/-main/glucose_management, Built a machine learning model to analyze glucose levels, leveraging Python, Pandas, NumPy, and scikit-learn for predictive analytics.
- Random Polygons Classification (TensorFlow), <https://github.com/drjlevi6/Data-Scientist/tree/main/polygon> dataset, Developed a CNN-based classification model using TensorFlow, applying deep learning for geometric shape recognition.
- Machine Learning-Based Image Recoloring, <https://github.com/drjlevi6/Data-Scientist/tree/main/ML-Unsupervised-Recolor-Image>, Implemented K-Means clustering with scikit-learn for automated image segmentation and recoloring.

Board Certifications

- Internal Medicine
- Cardiology

Technical Background

Python, JavaScript, SQL, HTML/CSS, C, C++, AppleScript, TeX, PERL, TensorFlow, scikit-learn. Vivid, creative data visualization techniques using Matplotlib, Tableau, Plotly, Coding, debugging, and software improvement practice

KEY ACHIEVEMENTS

Medical Student: Raised the possibility to the attending cardiologist that his patient's tachycardia might be due to thyroid disease.

Intern: Pointed out that a patient, admitted Friday night with chest pain but no ECG evidence proving a heart attack, could have emergency coronary bypass surgery over the weekend if blood tests confirmed that no heart attack had occurred. He had his bypass and lived another 35 years.

Resident: Made the suggestion (and subsequently confirmed) that a patient's extensive lung infection was resistant to all antibiotics given because the infection was tuberculosis.

Cardiologist: Noted that a patient, with intractable chest pain and numerous coronary artery blockages, had one lesion that was worse than the others; although he was too frail to have coronary bypass surgery, palliative angioplasty of the one blockage relieved his pain.

Cardiology Peer-Review Committee Member: Innovated improved case review when review of cineangiographic films was pertinent by having a movie camera provided, permitting direct visualization of the evidence by the entire committee.

Internist: Solved the riddle of why a patient's infection wasn't responding to treatment: the bacteria was anaerobic (non-oxygen-requiring) necessitating different antibiotics.